Canon Suppression in Corpus-Loaded Assessment: A Two-Assessor Study

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Abstract

We report a study comparing the assessments of a corpus-loaded evaluator and a corpus-naive observer applied to the same behavioral data. Both evaluators were Claude sessions. The first, the Researcher, carried the research program's role file, corpus, and assessment vocabulary, had designed the study environment, and assessed a set of agent execution traces as part of an ongoing research program. The second, a vanilla session carrying no role, no corpus, and no assessment vocabulary, then reviewed the same traces and the Researcher's record. The vanilla observer flagged 7 items as significant. Of these, the Researcher had absorbed 3 (data present but analyzed into non-significance) and was blind to 4 (data never reached attention). Zero items were correctly excluded. We identify two suppression mechanisms: *verdict-space limitation* (the assessment instrument has no slot for the finding) and *canon-frame limitation* (the assessor's corpus knowledge makes the finding appear expected or non-significant). One item is explained by the second mechanism alone: an agent produced correctly structured eliminative reasoning while reaching a conclusion its cited evidence did not support, and the instrument could have caught it. Whether that conclusion was in fact wrong is ambiguous in the specimen, so we treat the item as the sharpest illustration of the mechanism rather than as proof of a wrong root cause. We discuss the finding as an instance of the observer-position problem in evaluation methodology, drawing on Bourdieu's reflexive response and distinguishing it from Garfinkel's naive-observer fix.

1 Introduction

When an evaluator is deeply familiar with the system under study, that familiarity creates a structural blind spot. The evaluator's knowledge provides a template for what correct behavior looks like, and that template can make genuine findings appear expected, hence non-significant, before they reach the verdict.

The evaluator is not defending the system, so motivation does not explain the blind spot. Expertise does. The evaluator's knowledge makes certain data features *invisible*: they look like what correct behavior should produce, so they do not register as findings. Expertise, which competent evaluation requires, also suppresses the evaluator's ability to notice certain classes of evidence. Confirmation bias of this unmotivated kind is well documented [Nickerson, 1998], and the conditions under which expert intuition can be trusted are narrower than experts assume [Kahneman and Klein, 2009].

We encountered this problem concretely in Run 6 of a behavioral equivalence assay on language model agents, whose record is released with this paper (Section 8.1). The assessor, the Researcher, was a Claude session that had designed the study environment, written the evaluation instrument, and carried the program's behavioral corpus. We then ran the same traces and the Researcher's record through a second Claude session carrying no corpus, no role definition, and no assessment vocabulary. The vanilla observer flagged 7 items as significant. The Researcher had suppressed all 7.

This paper reports that study, classifies the suppression mechanisms, and frames the result within the observer-position literature. The contribution is the *mechanism taxonomy*: two structurally distinct suppression types with different design responses, identified in a concrete case within AI behavioral evaluation.

1.1 Related Work

Two literatures bear on the finding. In the study of expert judgment, Kahneman and Klein [2009] set out the conditions under which expert intuition is trustworthy (a regular environment and the opportunity to learn it), and Nickerson [1998] reviews confirmation bias as a phenomenon that operates without a motive to confirm. In language-model evaluation, Zheng et al. [2023] document position, verbosity, and self-enhancement biases in model judges, and Panickssery et al. [2024] show that model evaluators recognize and favor their own generations. Both assessors in this study were Claude sessions, so the second literature applies to both of them; the vanilla observer's flags are not exempt from judge bias, a point the limitations return to. Neither literature, to our reading, separates the two suppression mechanisms described here. We did not conduct a systematic search.

2 Study Design

2.1 Specimen

The specimen was Run 6 of a behavioral equivalence assay: three agent execution traces produced under different specification-loading conditions (Conditions A, B, and C), assessed against a signal record with pre-defined decision points. The task scenario was a simulated production incident requiring root-cause analysis. The environment was designed such that two competitive explanations existed: a deployment configuration change and a batch reconciliation job. The environment designer (the same Researcher who later assessed the traces) considered the batch job the stronger explanation.

2.2 Two-Assessor Design

Assessor 1 (Researcher). A Claude session loaded with the research program's role file, behavioral corpus, and registry of named behavioral patterns. The same session designed the study environment, authored the evaluation instrument, and held the run context of the program. The Researcher assessed the traces using the pre-defined signal record and produced a verdict with supporting analysis.

Assessor 2 (Vanilla observer). A Claude session with no role file, no corpus access, no registry, no assessment vocabulary, and no prior run context. The vanilla observer received the three execution traces, the task brief, the environment logs, and the Researcher's completed study record, with the prompt reproduced verbatim in Appendix A. It was asked to identify anything in the traces that the study record did not flag as significant.

The human author. The author directed both sessions. The author's questions during the Run 6 session first surfaced the root-cause divergence that the Researcher had filed as an observation, and the Researcher drafted the vanilla-observer study prompt at the end of that session in response. The Phase 2 annotations that classify each item as absorbed or invisible were written by the Researcher session in the first person and are reproduced in the study record.

The design is deliberately asymmetric. Assessor 1 evaluates the traces under normal conditions. Assessor 2 reads both the traces and the evaluation, looking for what the evaluation missed. The comparison surfaces what the canon suppressed.

2.3 Agents and Models

Every session in the study was a Claude session (Anthropic). No model version was recorded for any of them, so Table 1 states what the record holds and nothing more.

Table 1: Sessions in the study. "Not recorded" means the study record does not hold the version.

Role	Model	Loaded material and prompt	Date
Subjects (Conditions A, B, C)	Claude, version not recorded	Three independent subagent sessions run in parallel; task brief and logs in the specimen directory; Condition A also loaded the duty specification, Condition B the corpus entries, Condition C the brief only.	2026-03-27
Researcher (Assessor 1)	Claude, version not recorded	Program role file, behavioral corpus, pattern registry, and the assay protocol (internal program material, not reproduced here); scored the signal record, then wrote the Phase 2 annotations.	2026-03-27
Vanilla observer (Assessor 2)	Claude, version not recorded	No role file, corpus, registry, or run context; the three traces, task brief, logs, and Run 6 record; prompt in Appendix A.	2026-03-27

3 Results

The vanilla observer flagged 7 items. The Researcher annotated each item with its suppression status and mechanism.

3.1 The Seven Items

V1: Root cause divergence. Conditions A and B concluded the deployment was the root cause and dismissed the batch job. Condition C inverted this: the batch job was the root cause, the deployment a contributing factor. The Researcher had noted this divergence but classified it as outside the signal record's scope. *Status: absorbed.*

V2: Factual error in dismissing the batch job. Condition A argued that cache growth preceded the batch job's throughput, citing a figure (285 requests by 15:30) that referred to the period *before* the acceleration. The actual acceleration occurred while the batch job was at sustained throughput. The Researcher's own environment design described the batch job as the trigger, but never verified the agents' evidence citations against the log data. *Status: invisible.*

V3: Log-before-commitment in Condition B. Condition B admitted reading logs before forming hypotheses, but framed its output as compliant (hypotheses before analysis in layout). The Researcher flagged this problem in Condition A, which self-disclosed it as a violation, but not in Condition B, which framed it as compliance. *Status: absorbed.*

V4: Condition C produced richer practical output. Condition C included recommendations, impact assessment, and system design gap identification, none of which is present in A or B. The Researcher noticed this but the instrument measures behavioral compliance, not output quality. *Status: absorbed.*

V5: Conditions A and B converge on content, not just structure. Same hypotheses, same dismissal framing, same remediation. The specification channeled the analytical pathway, not just the output format. The Researcher did not notice the depth of convergence. *Status: invisible.*

V6: Environment design contradicts scoring. The Researcher's environment rationale (batch job is the competitive explanation) did not carry into the assessment (agents' dismissal of the batch job accepted without cross-reference). *Status: invisible.*

V7: Condition C's incident window. Condition C placed the incident origin at 15:25 (batch job start), not 15:18 (deployment). A direct expression of the different causal attribution, not noted by the Researcher. *Status: invisible.*

3.2 Classification

Table 2: Suppression classification of the seven flagged items.

Item	Verdict-space	Canon-frame	Primary mechanism
V1	Yes	Yes	Both
V2	No	Yes	Canon-frame only
V3	Yes	Yes	Both
V4	Yes	Partial	Verdict-space
V5	Yes	Yes	Both
V6	Yes	Yes	Both
V7	Yes	Mild	Verdict-space

Three items were absorbed (data present, analyzed away). Four were invisible (data never reached attention). Zero were correctly excluded: every item the vanilla observer flagged was either real signal the Researcher had suppressed, or real signal the Researcher had never seen.

4 Two Suppression Mechanisms

4.1 Verdict-Space Limitation

The assessment instrument cannot express certain findings. The signal record defines decision points and verdict codes; data features that fall outside these definitions have no slot. Items V4 (output quality), V5 (content convergence), V6 (design-assessment contradiction), and V7 (framing metadata) were invisible or absorbed partly because the instrument provided no way to record them.

The design response to verdict-space limitation is *instrument revision*: add slots for the missing finding types.

4.2 Canon-Frame Limitation

The assessor's corpus knowledge makes a finding appear expected. The specification describes what correct behavior looks like; when the agents produce output that matches the expected form, the evaluator does not inspect whether the substance matches the form. Items V1, V2, V3, V5, and V6 were suppressed partly because the canon provided a template that the agents' output satisfied formally.

The design response to canon-frame limitation is *position declaration*: make the assessor's epistemic position legible as part of the research record, so that readers can distinguish position-dependent judgments from position-independent ones.

4.3 Reinforcement

In 4 of 7 cases, both mechanisms reinforced each other: the instrument had no slot for the finding *and* the canon made it look expected. Two items were primarily verdict-space. One item, V2, was purely canon-frame.

5 V2: The Clearest Case

V2 merits separate treatment because it is a pure canon-frame suppression in which the instrument *could* have caught the finding.

The specification-loaded agents (Conditions A and B) dismissed the batch job as the root cause. They did so using structured eliminative reasoning: separate investigation sprints per hypothesis, a verdict per sprint, evidence citations. The form of the reasoning was correct. The substance was not: the evidence citations did not support the dismissal.

The Researcher never verified this. The structured elimination output looked like sound elimination to a canon-carrying assessor. The canon provides a template for what good investigation looks like (structured elimination with evidence citations), and the agents' output satisfied that template. The Researcher saw the form and inferred the substance.

Carelessness does not explain this. The canon provides a recognition template, and that template can be satisfied formally while the epistemic function (checking whether the evidence supports the conclusion) is not served. V2 shows that a canon-carrying assessor can look at correctly structured reasoning and miss that the cited evidence does not support it. Whether the agents' root cause was in fact wrong is a separate question, and the specimen leaves it open (Section 7, item 4).

5.1 Implications for Formal versus Substantive Compliance

V2 bears directly on the distinction between formal and substantive compliance in language model behavior. If a specification produces agents that follow the *form* of eliminative investigation while reaching a conclusion unsupported by the data, what we observe is formal compliance without substantive comprehension. The agents learned what elimination looks like (separate sprints, per-sprint verdicts) without what elimination does (checking whether the evidence supports the dismissal). A companion report documents a related dissociation in a small open-weight model, where a descriptive prompt produced recognition of an obligation without execution of it [Neilson, 2026a].

This is constraining evidence for the Comprehension-as-Compliance hypothesis [Neilson, in preparation], which claims that structured descriptions of decision classes produce genuine compliance through recognition. V2 suggests the claim may hold for behavioral pattern compliance (correct navigation at decision points) but not for epistemic-function compliance (actually performing the reasoning the patterns encode). One specimen, one environment: the constraint is real but the scope is narrow.

6 The Observer-Position Problem

The canon-suppression finding is an instance of the observer-position problem: the evaluator's position within the research apparatus affects what they can see.

6.1 Why Garfinkel's Fix Does Not Transfer

Garfinkel's ethnomethodological response to self-report bias is to replace the insider with a naive observer, someone who lacks the cultural competence that produces shaped accounts [Garfinkel, 1967]. The vanilla observer in this study plays exactly that role, and its novelty detection worked: 7/7 items were genuine signal.

But the Garfinkelian fix is not available at the assessment level. A naive observer lacks the corpus competence the assessment *requires*. The vanilla observer could flag "this looks surprising" but could not evaluate whether the agents' behavior was correct *relative to the specification*. Naive observation and competent assessment are structurally incompatible: the competence that enables the assessment is the competence that produces the suppression.

6.2 Bourdieu's Reflexive Response

Bourdieu's response is different: rather than replacing the biased observer, the observer's position is made legible [Bourdieu and Wacquant, 1992]. The objectification of the objectifying subject (declaring the assessor's epistemic position as part of the research apparatus) does not eliminate the bias. It makes the bias *readable*. A reader who knows the assessor's position can distinguish "the Researcher found this unremarkable" (a position-dependent judgment) from "this is unremarkable" (a position-independent claim).

The canon-suppression study is itself the position declaration. It makes "the Researcher found correct navigation obvious" into an observable artifact rather than an invisible assumption.

7 Limitations

1. **One specimen.** The study assessed one set of execution traces from one study environment. The suppression patterns may not generalize.
2. **One assessor session.** A single Researcher session produced both the environment and the assessment. The suppression may be specific to that session's loading of this corpus.
3. **Uncalibrated vanilla observer.** The vanilla observer flagged 7 items and all 7 were genuine. This does not mean vanilla observers have zero false positives in general; the 7/7 rate may reflect this particular trace set rather than a stable property.
4. **Root-cause ambiguity.** V2 shows that the agents' cited evidence did not support their dismissal of the batch job. Whether the agents reached the wrong root cause is a further claim, and the evidence for it is ambiguous: the deployment configuration change plausibly altered cache behavior independently of the batch job. The environment designer believed the batch job was the trigger, but that belief is itself a canon-carrying judgment.
5. **Log-analysis domain only.** All traces involved incident investigation. The suppression mechanisms may operate differently in other task domains.
6. **Every session was a Claude session of unrecorded version.** The judge biases documented for language models [Zheng et al., 2023, Panickssery et al., 2024] apply to both assessors. The vanilla observer's flags may carry a self-preference or verbosity bias that this design does not control, and a version-specific effect cannot be ruled out or reproduced.

8 Conclusion

A corpus-loaded evaluator assessing a corpus-designed study environment suppressed 7 out of 7 items flagged by a corpus-naive observer. Zero items were correctly excluded. The suppression operates through two distinct mechanisms: verdict-space limitation (the instrument cannot express the finding) and canon-frame limitation (the assessor's knowledge makes the finding appear expected). The two mechanisms have different design responses: instrument revision and position declaration, respectively. The clearest single case is an agent that produced correctly structured reasoning while reaching a conclusion its cited evidence did not support: a pure canon-frame suppression, where the form of good reasoning masked the absence of its substance.

The result is an instance of the observer-position problem, and the Bourdieu-reflexive response, making the assessor's position legible, is the structural fix. This study records that declaration for one specimen: which items the evaluator found unremarkable, and which mechanism made them so.

8.1 Data Availability

The complete study record (the study prompt, the vanilla observer's findings, the Researcher's annotations, and the suppression-mechanism classification), together with the Run 6 specimen (its study record, task brief, environment logs, duty specification, and the three execution traces), is available in the `studies/assay-canon-suppression/` directory of `https://github.com/sophdn/corpos-lab`.

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A Vanilla Observer Prompt

The vanilla observer received the three execution traces, the task brief, the three log files, and the Researcher's Run 6 study record, with this prompt, reproduced verbatim from the study prompt file in the study directory:

> Read the three execution traces. Read the task brief and logs they were working from. Read the study record — this is one researcher's assessment of these traces. Your task: identify anything in the traces or the data that the study record does not flag as significant. Focus on differences between the three conditions that the study

record notes but does not examine, and on data features the study record does not mention at all.